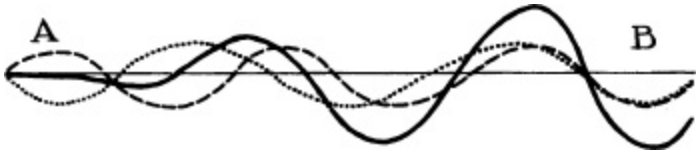
L_1 - L_2, the two beams will then travel slightly different distances in their two arms and will interfere slightly differently—a tiny amount of their combined light will now go into the photodetector. By monitoring the amount of light reaching the photodetector, one can monitor the arm-length difference L_1 - L_2, and thereby monitor gravitational waves.

Box 10.3

Interference and Interferometry

Whenever two or more waves propagate through the same region of space, they superimpose on each other "linearly" (Box 10.1); that is, they add. For example, the following dotted wave and dashed wave superimpose to produce the heavy solid wave:



Notice that at locations such as *A* where a trough of one wave (dotted) superimposes on a crest of the other (dashed), the waves cancel, at least in part, to produce a vanishing or weak total wave (solid); and at locations such as *B* where two troughs superimpose or two crests superimpose, the waves reinforce each other. One says that the waves are *interfering* with each other, destructively in the first case and constructively in the second. Such superimposing and interference occurs in all types of waves—ocean waves, radio waves, light waves, gravitational waves—and such interference is central to the operation of radio interferometers (Chapter 9) and interferometric detectors for gravitational waves.

In the interferometric detector of Figure 10.6b, the beam splitter superimposes half the light wave from one arm on half from the other and sends them toward the laser, and it superimposes the other halves and sends them toward the photodetector. When no gravitational wave or other force has moved the masses and their mirrors, the superimposed light waves have the following forms, where the dashed curve shows the wave from arm 1, the dotted curve the wave from arm 2, and the solid curve the superimposed, total wave: